

The Big Bang: 0/0 of Origin

Michael Grafiel S Puno

2026

The Removable Singularity Framework

Abstract

We show that the Big Bang IS a 0/0 removable singularity: the origin of everything. At $T = 0$, density and temperature are infinite (undefined). Inflation expanded the universe by 10^{26} in 10^{-32} seconds, solving the horizon and flatness problems. $\Omega = 1.000$ (exactly!) is explained by inflation. The matter-antimatter asymmetry ($\eta = 6e-10$) created all matter. Quantum gravity may resolve the singularity. This connects to the cosmic web, arrow of time, holographic principle, measurement problem, and black holes.

1. The Singularity (Cosmic 0/0)

$T = 0$: density = infinity, temperature = infinity

At $T = 0$, the universe was a singularity: density and temperature are infinite (undefined). This is the 0/0 of origin. No "before" Big Bang: time begins at $T = 0$. Quantum gravity may resolve this singularity (Loop quantum cosmology: Big Bounce).

2. Inflation

Factor = $\exp(60) \sim 10^{26}$ in 10^{-32} seconds

Inflation expanded the universe by a factor of 10^{26} in 10^{-32} seconds. This exponential expansion solves the horizon problem (uniform CMB) and flatness problem ($\Omega = 1$). The universe grew from smaller than a proton to larger than a galaxy in 10^{-32} seconds.

3. Horizon Problem

The CMB is uniform in ALL directions, but regions are too far apart to have ever been in contact. Inflation solves this: regions WERE in contact before inflation, then were separated. This is the 0/0: uniformity from chaos.

4. Flatness Problem

$\Omega = 1.000$ (exactly!)

The universe is exactly flat ($\Omega = 1$). Inflation drives Ω to 1 exponentially. Any initial value of Ω becomes 1 after 60 e-folds of inflation. This explains why the universe is flat.

5. Matter-Antimatter Asymmetry

$\eta = (n_b - n_{\bar{b}}) / n_{\gamma} \sim 6e-10$

For every 10^9 antimatter particles, there were $10^9 + 6$ matter particles. This tiny asymmetry (6 parts per billion) created ALL matter in the universe. CP violation in weak interactions is the mechanism.

6. Connections

System

Prior Chapter

Connection

Cosmic web	Ch.46 Cosmic web	<i>Puno (2026) - The Big Bang: 0/0 of Origin</i> Structure formation
Arrow of time	Ch.48 Arrow of time	Past Hypothesis
Holographic	Ch.47 Holographic	Cosmological info
Measurement	Ch.49 Measurement	The observer
Black holes	Ch.32 Black holes	Singularities
SOC	Ch.41 SOC	Self-organization

7. Conclusion

The Big Bang IS a 0/0: the origin of everything. Inflation expanded the universe by 10^{26} in 10^{-32} seconds, solving the horizon and flatness problems. $\Omega = 1.000$ is explained by inflation. The matter-antimatter asymmetry created all matter. Quantum gravity may resolve the singularity. This is the LARGEST 0/0 in existence, connecting the cosmic web, arrow of time, holographic principle, measurement problem, and black holes.

References

- [1] Guth (1981), inflationary universe
- [2] Penzias & Wilson (1965), CMB discovery
- [3] Planck Collaboration (2020), cosmological parameters
- [4] Sakharov (1967), matter-antimatter asymmetry
- [5] Puno, "The Removable Singularity" (2026)